# FCC TEST REPORT

| | |
|---|---|
| **Report Number**.....: | **ZKT-2601161131E** |
| Date of Test..... | Jan. 15, 2026 to Jan. 22, 2026 |
| Date of issue.....: | Jan. 22, 2026 |
| Total number of pages..... | 12 |
| Test Result ..... : | PASS |
| **Testing Laboratory**.....: | **Shenzhen ZKT Technology Co., Ltd.** |
| Address ..... : | 1/F, No. 101, Building B, No. 6, Tangwei Community Industrial Avenue, Fuhai Street, Bao'an District, Shenzhen, China |
| **Applicant's name .....:** | **Guangzhou Xingyi Electronic Technology Co., Ltd.** |
| Address ..... : | 4th & 16th Floors, Building 2, Yudong Sci-Tech Park, No.3 Ketai 1st Road, Baiyun District, Guangzhou. |
| **Manufacturer's name .....:** | **Guangzhou Xingyi Electronic Technology Co., Ltd.** |
| Address ..... : | 4th & 16th Floors, Building 2, Yudong Sci-Tech Park, No.3 Ketai 1st Road, Baiyun District, Guangzhou. |
| **Test specification:** | |
| Standard..... : | FCC Part 15 B<br>ANSI C63.4:2014 |
| Test procedure.....: | FCC |
| Non-standard test method .....: | N/A |
| **Test Report Form No..... :** | TRF-EL-117_V0 |
| **Test Report Form(s) Originator..... :** | ZKT Testing |
| **Master TRF .....:** | Dated: 2020-01-06 |

This device described above has been tested by ZKT, and the test results show that the equipment under test (EUT) is in compliance with the FCC requirements. And it is applicable only to the tested sample identified in the report.
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| | |
|---|---|
| **Product name..... :** | **ATK-CLRK3562JB CORE BOARD** |
| Trademark .....: | N/A |
| Model/Type reference..... : | ATK-CLRK3562JB |
| Ratings..... : | Input: DC 12V1A |

**Testing procedure and testing location:**

| | |
|---|---|
| **Testing Laboratory.....:** | **Shenzhen ZKT Technology Co., Ltd.** |
| Address.....: | 1/F, No. 101, Building B, No. 6, Tangwei Community Industrial Avenue, Fuhai Street, Bao'an District, Shenzhen, China |
| Tested by (name + signature).....: | Jim Liu |
| Reviewer (name + signature).....: | Jackson Fang |
| Approved (name + signature).....: | Lake Xie |

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## 1.VERSION

| Report No. | Version | Description | Approved |
|---|---|---|---|
| ZKT-2601161131E | Rev.01 | Initial issue of report | Jan. 22, 2026 |
| | | | |
| | | | |

## 2. GENERAL INFORMATION

### 2.1 Description of Device (EUT)

| | | |
|---|---|---|
| EUT | : | ATK-CLRK3562JB CORE BOARD |
| Trademark | : | N/A |
| Model Number | : | ATK-CLRK3562JB |
| Model Difference | : | N/A |
| Power Supply | : | Input: DC 12V1A |
| The highest frequency of the internal sources of the EUT is (less than 108)MHz: | | ☒ less than 108 MHz, the measurement shall only be made up to 1 GHz.<br>☐ between 108 MHz and 500 MHz, the measurement shall only be made up to 2 GHz.<br>☐ between 500 MHz and 1 GHz, the measurement shall only be made up to 5 GHz.<br>☐ above 40 GHz, the measurement shall be made up to 5 times the highest frequency or 40 GHz, whichever is less. |

Note: N/A

### 2.2 Tested System Details

None.

### 2.3 Test Facility

Shenzhen ZKT Technology Co., Ltd.
Add. : 1/F, No. 101, Building B, No. 6, Tangwei Community Industrial Avenue, Fuhai Street, Bao'an District, Shenzhen, China

FCC Test Firm Registration Number: 692225
Designation Number: CN1299
IC Registered No.: 27033

### 2.4 MEASUREMENT UNCERTAINTY

Where relevant, the following measurement uncertainty levels have been estimated for tests performed on the Product as specified in CISPR 16-4-2. This uncertainty represents an expanded uncertainty expressed at approximately the 95% confidence level using a coverage factor of k=2.

| Test item | Value (dB) |
|---|---|
| Conducted Emission (150K-30MHZ) | 3.20 |
| Radiated disturbance30MHz-1000MHz | 4.80 |
| | |

## 2.5 Test Instrument Used

### Radiation Emissions Test

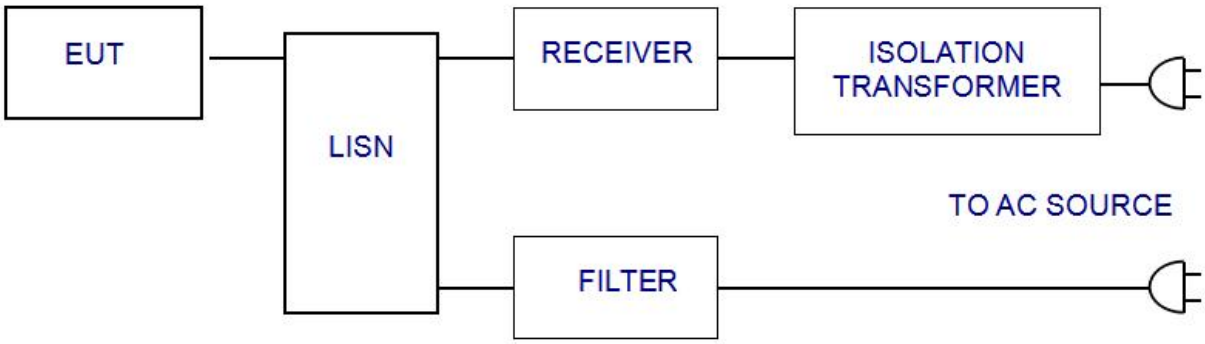
| Item | Equipment | Manufacturer | Type No. | Serial No. | Firmware Version | Last calibration | Calibrated until |
|---|---|---|---|---|---|---|---|
| 1 | Spectrum Analyzer (9kHz-26.5GHz) | KEYSIGHT | N9020A | MY55370835 | A.17.05 | Sep. 20, 2025 | Sep. 19, 2026 |
| 2 | Spectrum Analyzer (10kHz-39.9GHz) | R&S | FSV40-N | 100363 | 1.71 SP2 | Sep. 20, 2025 | Sep. 19, 2026 |
| 3 | EMI Test Receiver (9kHz-7GHz) | R&S | ESCI7 | 100969 | 4.32 | Oct. 24, 2025 | Oct. 23, 2026 |
| 4 | Bilog Antenna (30MHz-1500MHz) | Schwarzbeck | VULB9168 | 00877 | N/A | Sep. 20, 2025 | Sep. 19, 2026 |
| 5 | Horn Antenna (1GHz-18GHz) | Agilent | AH-118 | 071145 | N/A | Sep. 20, 2025 | Sep. 19, 2026 |
| 6 | Horn Antenna (15GHz-40GHz) | A.H.System | SAS-574 | 588 | N/A | Sep. 30, 2025 | Sep. 29, 2026 |
| 7 | Loop Antenna | TESEQ | HLA6121 | 58357 | N/A | Sep. 20, 2025 | Sep. 19, 2026 |
| 8 | Amplifier (30-1000MHz) | EM Electronics | EM330 Amplifier | 60747 | N/A | Sep. 19, 2025 | Sep. 18, 2026 |
| 9 | Amplifier (1GHz-26.5GHz) | HuiPu | 8449B | 3008A00315 | N/A | Sep. 19, 2025 | Sep. 18, 2026 |
| 10 | Amplifier (500MHz-40GHz) | QuanJuDa | DLE-161 | 097 | N/A | Oct. 24, 2025 | Oct. 23, 2026 |
| 11 | Test Cable | N/A | R-01 | N/A | N/A | May. 25, 2025 | May. 24, 2026 |
| 12 | Test Cable | N/A | R-02 | N/A | N/A | May. 25, 2025 | May. 24, 2026 |
| 13 | Test Cable | N/A | R-03 | N/A | N/A | May. 25, 2025 | May. 24, 2026 |
| 14 | D.C. Power Supply | LongWei | TPR-6405D | N/A | N/A | \ | \ |
| 15 | EMC Software | Frad | EZ-EMC | Ver.EMC-CON 3A1.1 | N/A | \ | \ |
| 16 | Turntable | MF | MF-7802BS | N/A | N/A | \ | \ |
| 17 | Antenna tower | MF | MF-7802BS | N/A | N/A | \ | \ |

### Conduction Emissions Test

| Item | Kind of Equipment | Manufacturer | Type No. | Serial No. | Firmware Version | Last calibration | Calibrated until |
|---|---|---|---|---|---|---|---|
| 1 | LISN | R&S | ENV216 | 101471 | N/A | Sep. 23, 2025 | Sep. 22, 2026 |
| 2 | LISN | CYBERTEK | EM5040A | E1850400149 | N/A | Sep. 19, 2025 | Sep. 18, 2026 |
| 3 | Test Cable | N/A | C-01 | N/A | N/A | May. 25, 2025 | May. 24, 2026 |
| 4 | Test Cable | N/A | C-02 | N/A | N/A | May. 25, 2025 | May. 24, 2026 |
| 5 | Test Cable | N/A | C-03 | N/A | N/A | May. 25, 2025 | May. 24, 2026 |
| 6 | EMI Test Receiver | R&S | ESCI3 | 101393 | 4.42 SP3 | Oct. 24, 2025 | Oct. 23, 2026 |
| 7 | Triple-Loop Antenna | N/A | RF300 | 9194 | N/A | Sep. 19, 2025 | Sep. 18, 2026 |
| 8 | Absorbing Clamp | DZ | ZN23201 | 15034 | N/A | Sep. 23, 2025 | Sep. 22, 2026 |
| 9 | EMC Software | Frad | EZ-EMC | Ver.EMC-CON 3A1.1 | N/A | \ | \ |

## 3.CONDUCTED EMISSION AT THE MAINS TERMINALS TEST

### 3.1 Block Diagram Of Test Setup




### 3.2 Test Standard

FCC PART 15 B

### 3.3 Power Line Conducted Emission Limit

| Frequency MHz | Limits dB(μV) | |
|---|---|---|
| | Quasi-peak Level | Average Level |
| 0.15 ~ 0.50 | 66 ~ 56* | 56 ~ 46* |
| 0.50 ~ 5.00 | 56 | 46 |
| 5.00 ~ 30.00 | 60 | 50 |

Notes: 1. *Decreasing linearly with logarithm of frequency.
2. The lower limit shall apply at the transition frequencies.

### 3.4 EUT Configuration on Test

The following equipments are installed on conducted emission test to meet FCC PART 15 B requirement and operating in a manner which tends to maximize its emission characteristics in a normal application.

### 3.5 Operating Condition of EUT

3.5.1 Setup the EUT and simulators as shown in Section 3.1.
3.5.2 Turn on the power of all equipments.
3.5.3 Let the EUT work in test modes and test it.

### 3.6 Test Procedure

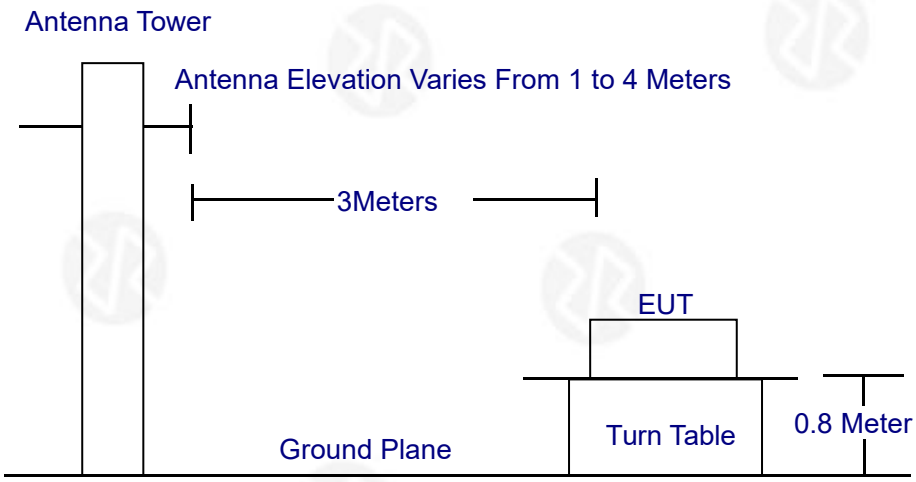
The EUT is put on the ground and connected to the AC mains through a Artificial Mains Network (AMN). This provided a 50ohm coupling impedance for the tested equipments. Both sides of AC line are checked to find out the maximum conducted emission levels according to the **FCC PART 15 B** regulations during conducted emission test.
The bandwidth of the test receiver (R&S Test Receiver ESCI) is set at 10KHz.
The frequency range from 150 KHz to 30 MHz is investigated.

### 3.7 Test Result

The EUT is powered by DC only the test items is not applicable

## 4.RADIATION EMISSION TEST

### 4.1 Block Diagram of Test Setup




### 4.2 Test Standard

FCC PART 15 B

### 4.3 Radiation Limit

| FREQUENCY (MHz) | DISTANCE (Meters) | FIELD STRENGTHS LIMITS (dBμV/m) |
|---|---|---|
| 30 ~ 88 | 3 | 40.0 |
| 88 ~ 216 | 3 | 43.5 |
| 216 ~ 960 | 3 | 46.0 |
| 960 ~ 1000 | 3 | 54.0 |

### 4.4 EUT Configuration on Test

The FCC PART 15 B regulations test method must be used to find the maximum emission during radiated emission test. The configuration of EUT is the same as used in conducted emission test.Please refer to Section 2.2.

### 4.5 Operating Condition of EUT

Same as conducted emission test, which is listed in Section 2.2 except the test set up replaced as Section 4.1.

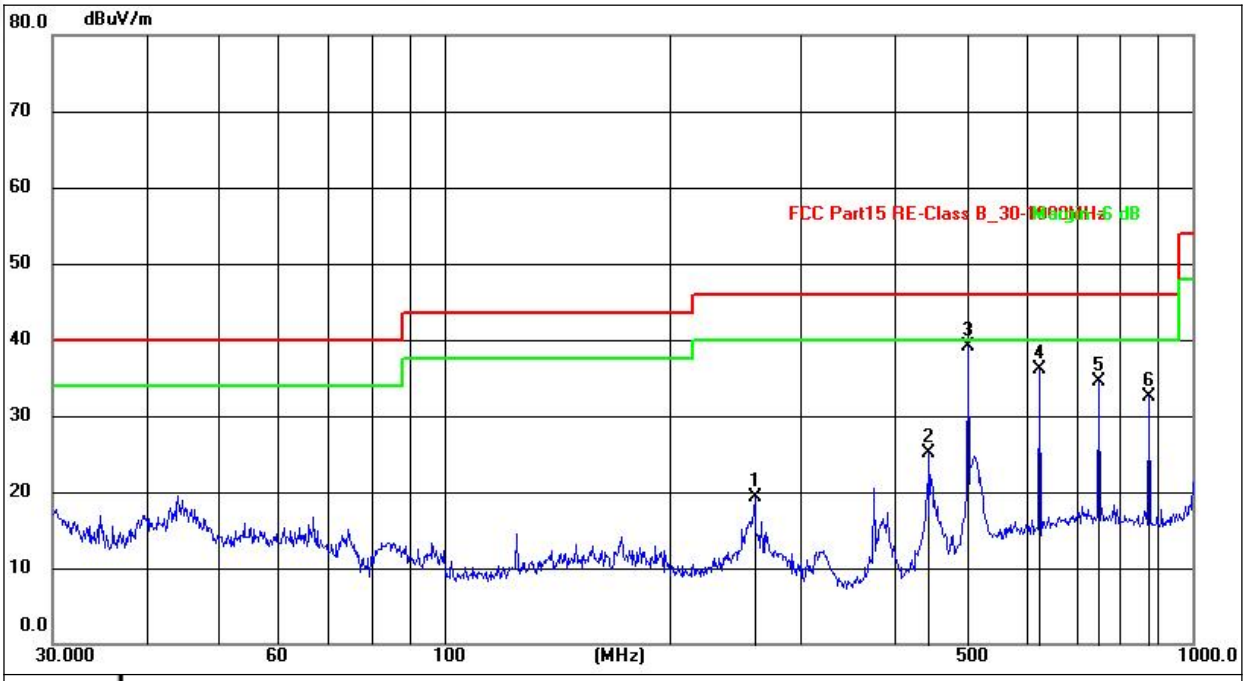
### 4.6 Test Procedure

The EUT and its simulators are placed on a turned table that is 0.8 meter above the ground. The turned table can rotate 360 degrees to determine the position of the maximum emission level. The EUT is set 3 meters away from the receiving antenna that is mounted on the antenna tower. The antenna can move up and down between 1 meter and 4 meters to find out the maximum emission level. Broadband antenna (calibrated biconical and log periodical antenna) is used as receiving antenna. Both horizontal and vertical polarization of the antenna is set on test. In order to find the maximum emission levels, the interface cable must be manipulated according to FCC PART 15 B on radiated emission test.
The bandwidth setting on the field strength meter (R&S Test Receiver ESCI) is set at 120KHz below 1GHz, set at 1MHz above 1GHz.The frequency range from 30MHz to 1000MHz is checked.The highest frequency of the internal sources of the EUT was below 108MHz, so the measurement was only made up to 1GHz.

### 4.7 Test Result

PASS
Please refer to the following page.

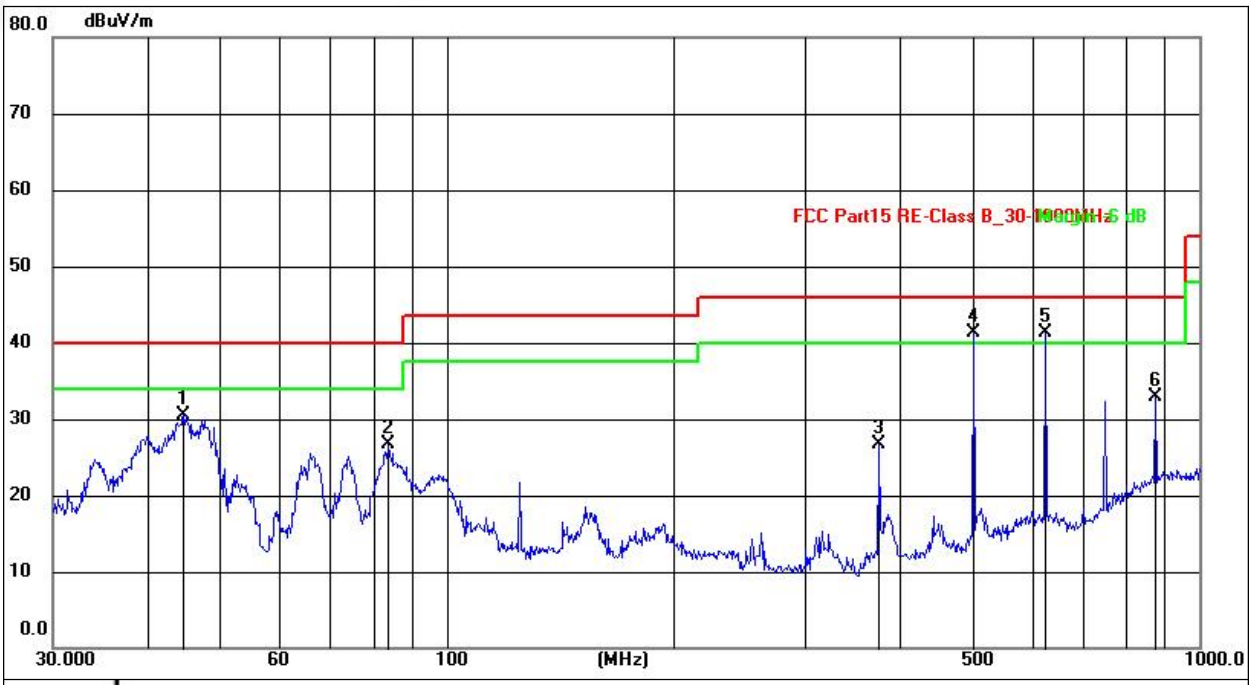
| Radiation Emission Test Data | | | |
|---|---|---|---|
| Temperature: | 26℃ | Relative Humidity: | 60% |
| Pressure: | 1009hPa | Phase : | Horizontal |
| Test Voltage : | DC 12V | Test Mode: | Working |




| No. | Frequency (MHz) | Reading (dBuV) | Factor (dB/m) | Level (dBuV/m) | Limit (dBuV/m) | Margin (dB) | Detector |
|---|---|---|---|---|---|---|---|
| 1 | 260.1444 | 38.79 | -19.55 | 19.24 | 46.00 | -26.76 | QP |
| 2 | 444.8514 | 44.30 | -19.29 | 25.01 | 46.00 | -20.99 | QP |
| 3 | 501.1788 | 55.85 | -16.67 | 39.18 | 46.00 | -6.82 | QP |
| 4 | 625.0778 | 49.97 | -13.80 | 36.17 | 46.00 | -9.83 | QP |
| 5 | 750.1082 | 47.26 | -12.76 | 34.50 | 46.00 | -11.50 | QP |
| 6 | 875.2468 | 45.90 | -13.32 | 32.58 | 46.00 | -13.42 | QP |

Notes:
1.An initial pre-scan was performed on the peak detector.
2.Quasi-Peak measurement were performed at the frequencies with maximized peak emission.
3.The emission levels of other frequencies are very lower than the limit and not show in test report.
4.Final Level = Reading level + Correct Factor
5.Correct Factor= Antenna factor+ Cable loss factor - Amplifier factor
6.Margin= Measurement Level-Limit

| Radiation Emission Test Data | | | |
|---|---|---|---|
| Temperature: | 26℃ | Relative Humidity: | 60% |
| Pressure: | 1009hPa | Phase : | Vertical |
| Test Voltage : | DC 12V | Test Mode: | Working |

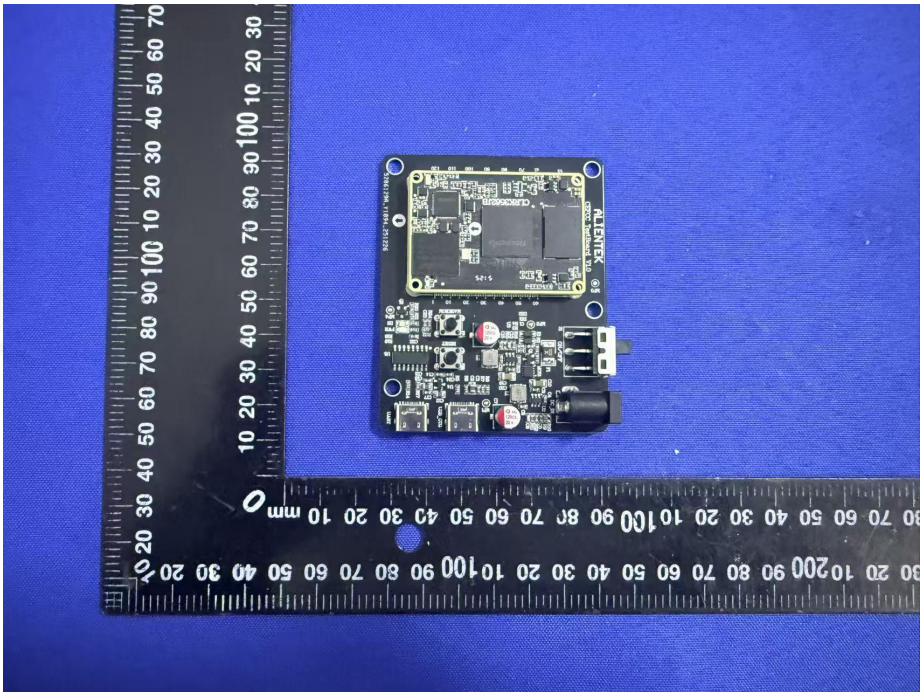



| No. | Frequency (MHz) | Reading (dBuV) | Factor (dB/m) | Level (dBuV/m) | Limit (dBuV/m) | Margin (dB) | Detector |
|---|---|---|---|---|---|---|---|
| 1 | 44.7433 | 48.79 | -18.34 | 30.45 | 40.00 | -9.55 | QP |
| 2 | 83.8155 | 49.86 | -23.24 | 26.62 | 40.00 | -13.38 | QP |
| 3 | 375.9384 | 47.30 | -20.59 | 26.71 | 46.00 | -19.29 | QP |
| 4 | 501.1788 | 57.84 | -16.61 | 41.23 | 46.00 | -4.77 | QP |
| 5 | 625.0778 | 54.65 | -13.25 | 41.40 | 46.00 | -4.60 | QP |
| 6 | 875.2468 | 40.60 | -7.78 | 32.82 | 46.00 | -13.18 | QP |

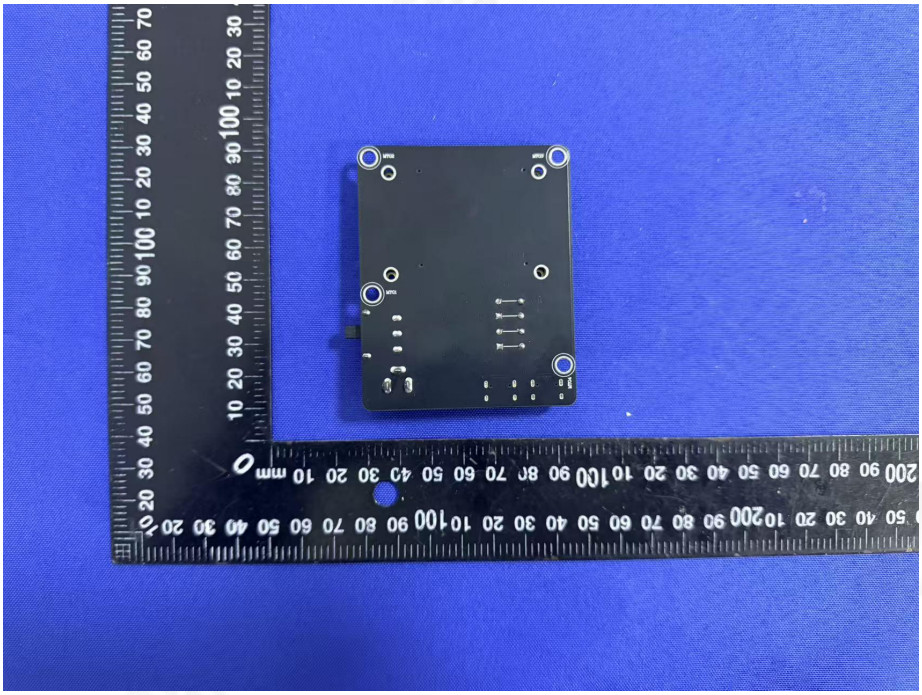
Notes:
1.An initial pre-scan was performed on the peak detector.
2.Quasi-Peak measurement were performed at the frequencies with maximized peak emission.
3.The emission levels of other frequencies are very lower than the limit and not show in test report.
4.Final Level = Reading level + Correct Factor
5.Correct Factor= Antenna factor+ Cable loss factor - Amplifier factor
6.Margin= Measurement Level-Limit

## 5.EUT PHOTOGRAPHS

### EUT Photo 1

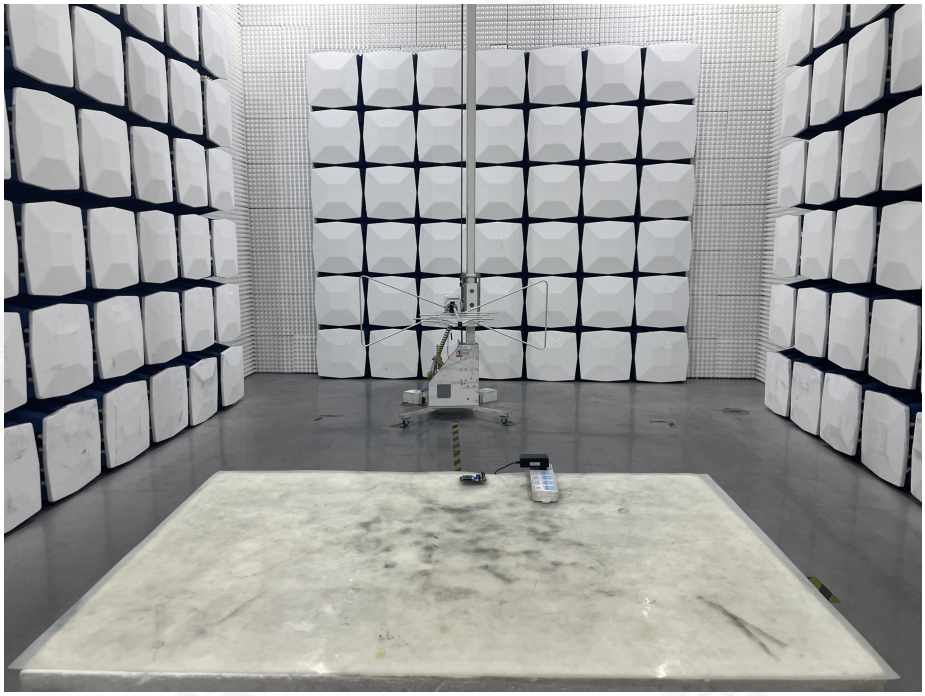


### EUT Photo 2



## 6.EUT TEST PHOTOGRAPHS

### RE



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